

ActInf Livestream #004.2: “Cultural Affordances: Scaffolding Local Worlds”

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Duration: 1:36:43 | **Uploaded:** Unknown **Transcript method:** auto_caption

all right i think we are live hello everyone and welcome to team com podcast 4.2 this is an exciting day because we are testing out new technology and so thanks to all of our participants we hope this works for all of you listening and in attendance let's go to the slides welcome to teamcomp teamcom is an experiment in online team communication and learning related to active inference you can find us on twitter at our inferenceactivehandle you can email us at activeinference gmail.com find us at our public key based team or watch us on youtube this is a recorded and archived live stream so please provide us with feedback so we can improve on our work all backgrounds and perspectives are welcome here we welcome all questions that can help people understand active inference and as far as basic video etiquette please mute if it's noisy in your background feel free to use the raise hand feature in jitsi so that we can make sure to call on you we are here in active inference stream 4.2 following up on 4.1 in 4.1 we discussed cultural affordances scaffolding local worlds through shared intentionality and regimes of attention a 2016 paper by ramstead at all today we're going to do a check-in and warm-up and last time we had a discussion of the daniel we can see slice by now yes okay good point i can see the slides on the live stream but uh they're not being shared in the jitsi so what i'm going to do and we're um doing it live i'm going to post the live stream link and if you'd like you could have the live stream muted in the background and just see what the slides are because it will just be rapidly um following so i'll continue on this first part and the link will be dream put into the jitsi chat if you could just give me one second how about something on active inference and multitasking that's what i want to know about last time we had an awesome discussion of the paper we went through the abstract the claims and the aims of the work and we also went through all the figures in pretty nice detail this time we are going to have further discussion which has been separated into a few different topics that none of the conversions have been prepared on beforehand so it will be fun and improvisational so let's start with the check-in and warm-up people can introduce themselves and their location if

they'd like the first question is what is a local world that you live in any thoughts on local worlds it's a good question um i mean i i you know i guess this says something about the way that we live i mean you know i i live in montreal and my local world uh i live around the mcgill area so it's very studenty um i guess i i live in a world where people share progressive values and i also get the feeling that there's something like a rift you know where where several of us are sharing the same geographical location um but don't necessarily live in the same world uh per se in the same in the sense of the things don't afford the same kinds of things that they do uh it you know for for other people living in another as it were world um yeah yep the first thing i thought of was that local can be geospatial or it can be informational one informational link away is just one video chat away one phone call away and what we're experiencing in a big way is that tension between our informational local worlds and our spatial ones and that can lead to um discordance any other thoughts on this part cool maybe uh in this term of local maybe it's based on communications which happens most often with you if even in a global environment but if you communicate each day with somebody your kind of local world cool second question what gives meaning to our local worlds so this can be maybe a thought from the paper what did they propose gives meaning to local worlds or just experientially what is it that gives meaning to your local world yeah stephen well i think there's the geographical like physical embeddedness of that world so how you actually when i think of world i kind of think very much of it in terms of the physical space being part of the of that whole piece um but then like you say how those things show up then starts to become maybe more abstract or more inferred through other people yep and from a process perspective as we'll get to later it's the feedback loops that happen within ourself and amongst various people that provide semiotic meaning sasha yeah for some reason i just thought of this phrase like theory of place kind of going along with theory of mind that in this kind of uh local information environment but maybe not geospatially local that the people that you're communicating with you may or may not know where they're communicating from and that um could be helpful in your communication or not because um simply because we're all here on this live stream we don't really need to know the exact location of each of the participants but if i did know it might add something to our communication or the narrative that we're building to kind of know where we can all connect from and then to follow up with the thinking through other minds idea it's like thinking through other spaces it's about putting yourself in someone else's shoes understanding where they're coming from these are embodied and spatial metaphors that we use just like can you grasp it can you sketch out the idea these

are tactile ways to talk about abstract things like ideas and so you're literally putting yourself in someone else's shoes if you ask what is the accessibility of this room going to be for somebody who's coming from a different place and these are the important things to consider and we'll return to this i think when we talk about the third space that exists in social interactions um in terms of the uh the paper cultural affordances the proposal there is something like well um what what's at stake is something like a dialectic between affordances and expectations right so uh what what ends up shaping the the worlds the that we inhabit right which which we're conceptualizing in terms of affordances um really are the the shared expectations that we have uh you know with with other people in our community and by shared expectations i mean something very broad uh to encompass uh the beliefs that we have about uh the shared world but also the norms uh that are kind of structuring uh our appraisal and interaction with the world as well as practices that might be ongoing and crucially the the shared physical environment that we have in common that in in the case of humans comes to encode information about the way that we live so you know think about uh the fact that in most cities streets meet at right angles uh or you know the use of circular uh traffic signalization uh to kind of coordinate our patterns of movement throughout a city i think what's at stake is really the this dialectic yep the ant colonies architecture is in feedback with their behavior and they're exactly updated and generated by the very same process which is nice construction or distributed cognition different contexts are going to have different terms uh final warm-up question is what are you trying to get a grip on this is looking forward to talking about grip later on in this discussion but what's just something that you're trying to get a grip on i just thought it'd be a fun random question one thing we were talking about just right before we started was trying to get a grip on some of these formalisms and understanding which formalisms in the free energy principle and in active inference are downstream of each other which ones are necessary or sufficient for each other and also what is the best way to on-ramp and to learn about them from various different perspectives i was going to say exactly that i was going to talk about the reading group and the progress that we're making there uh yeah what have you found interesting or liked about that well i mean i didn't expect this to really turn into the rich discussion and exchange community that it's turned into uh i mean uh yeah we're really coming to grips i think as a group with uh with the basically all the math and physics that goes into it like really uh becoming familiar with all the moving parts and all the ingredients it's been a really positive experience so that's what we're i guess i'm also trying to learn that guitar a bit more that's another thing i'm coming to literally grips with but those are

the two main things i'd say nice and stephen do you raise your hand yeah i like well like you were so talking about this idea of putting yourself or getting in someone else's shoes so i'm trying to like put myself or trying to understand to put myself in someone someone else's shoes in terms of understanding what i'm talking about this stuff and then i think the whole idea of like am i trying to understand um in this dialectic with someone else where they are in relation to me or am i actually getting into their shoes in a place or in a field of practice or in some sort of stance in relation to um some concepts so i think there's something kind of about actually how we try and understand what it means in a meta level like you know because and how we move between them because we i tend to find that a lot of um the stuff like anthropology and that it tends to have particular metaphors that we end up all having to follow from gold or whoever and that's particularly true in the arts it's all like it's all to lose and to lose and lose and then everything's those metaphors so like how how do you jump in and out of those different sort of shoes that people are in i think that's really interesting yep being specific with generalizations being tangible with the abstraction those are tensions that we work with and also one thought on local worlds and getting a grip so once there's a grip on the object it becomes part of your extended cognition if you have the grip on the hammer you can hit the nail but if you don't have a grip on the hammer it's not part of your affordances to nail that in with a hammer or if you're in someone else's the local world kind of becomes the foot shoe container and the local world is what gives meaning to that coupling between the foot and the shoe between the hand and the hammer and so in in just a spatial way but perhaps also in an abstract way the local world is tying back to the affordance idea and upon that control um idea sasha sasha muted oh uh yeah just to add a little uh you know meta level uh comment on that um the way that we communicate um right it has to capture the content of what we're trying to communicate and then also in a way that makes sense to uh the receiving party but it's just funny especially on this sort of podcast conversation style that um we're trying to communicate with each other but also in a way that is going to be understandable to a future audience and so i think there's a few different levels of modulating um our communication style and um the kind of affordances that we have perfect transition to our first bucket of questions which is feedback and affordances so on this one this slide has a picture of a termite nest it also has a picture of some brain computer images and there's a quote from the paper it reads this multi-level embodiment of the generative model as we shall argue below extends to the concrete material human design milieus or designer environments in which humans operate some generative models in this wide sense involve semantic content and others do not

they involve something more minimal than satisfaction conditions i.e reliable covariation so my question which is pretty broad was what does active inference tell us about niche construction in the physical sense or in the cultural sense there's a lot to say there and there were many subsequent papers um well i mean if i can just uh make a few points uh to get us started um so axel constant and yellow brinaberg i think are the two first people to really start working on the niche construction aspect of active inference back in 2017 or so and so all this work rests on the fact that whenever we're talking about active inference we're talking about a story that's symmetric in some deep sense so active inference is a story about how uh organisms over the course of uh you know development and uh and learning and through perception and processes like evolution become more like their ecological niche which technically means that their statistical structure comes to recapitulate uh in some sense the structure of the uh in which uh the you know the organisms that we're interested in exist um so for example uh you know you can tell a lot about my environment just by looking at my physical body right so the fact that i'm wearing a parka tells you that summer's over in canada for example so there are these kind of predictive relations between the state of my phenotype and the expected state of the environment to kind of come together so the other side of active inference that's often emphasizes the active bit so you know rather than making my beliefs more like the environment i can make my environment more like my beliefs or conformed to my preferred data distributions what was overlooked before 2017 is that from the point of view of the organism and environment the story is symmetric so when we're making the environment more like our preferred data as an active inference we're effectively changing the structure of enviro of the to to make it more like our own structure or to make it like the the like the structure of the the preferences that we're encoding um and so uh i think a lot of a lot of the more interesting work done on culture recently in the active inference tradition draws on uh this niche construction stuff i mean you know at some point the idea is you've only got so many layers of cortex and cortex cortex is metabolically expensive so where are you gonna keep optimizing you know after like the sixth layer or whatever of like increasingly meta increasingly centralized uh representation well one way to go is really just in the environment so there's a there's a sense in which really like coordinating amongst amongst con specifics you know like uh or or with a team member or whatever is sort of like constructing a niche in another person in a sense you're sort of like you know uploading into another person the sets of priors that will optimally allow you to interface together uh which relates to some of the top top effects that we were discussing last week so i mean ultimately the picture that i kind of

draw for multi-scale active inference is that you have two stacks essentially you've got a kind of vertical stack and a horizontal stack by vertical stack i mean that all the systems that we're considering are nested systems of systems of systems uh you know our bodies are made of organs that are made of cells and so on um or you know uh the broader uh socio-economic uh community in north america is made up of like you know more regional groups like the u.s canada mexico which in turn are made up of you know so you got these kind of nested uh structures going and the yeah the active inference story then can be used to i think make sense of the ways in which we structure the environment especially humans this is especially the case for humans that we structure the environment in order to foster adaptive patterns of group behavior any other thoughts on that and i can add as well well and raise your hand if so one part that i thought was really interesting maxwell was this notion of like a deep symmetry especially when the two interacting components of the system have very different affordances so i was thinking about winter just like you brought up we don't recapitulate uh snow or cold weathers however we do come to embody those regularities in the day and in the year with our first our thermoregulation as an organism and then with our behavior and with our extended phenotype with our buildings and with our air conditioners and so we do in a very deep sense come to embody those regularities and then the air conditioners change the climate in some sense maybe a little maybe a lot maybe it's in a good direction maybe it's in a bad direction but there is a consequence of this coming to embody the regularities and especially when the systems are very different maybe when you're doing a call and response it looks symmetrical at the syntactic level and it has this deep symmetry at the turn taking level and then with other systems where the where the turn taking looks very different it's still something symmetrical but it's not quite symmetrical at the first pass right well so the the uh the idea was i was trying uh elaborate a bit earlier with this vertical and horizontal stack is touches precisely on this is that it's not just that organisms are stacked in this kind of vertical ways that you also have this kind of horizontal relationship between an organism or an agent and its niche that turn out to also be stacked in this kind of way so it's not just that my cells make up organs that make up organisms that make up my body for instance it's that my body is a niche for or for my organs and my organs are a niche for my cells and so when we're talking about niche construction it's very much as sasha was describing a few a few moments ago it's that we're thinking through the environment that is populated by others we're thinking through other worlds uh by the way i noticed that my camera is very ours is not so you can reload one one thought i'll give there on the uh the primacy of the organism from a philosophy of biology or natural selection

perspective is it was seen that the organism or more specifically to some the genes of the organism are the only thing that make it through the weismann barrier the only thing to make it into the next generation is the information that encodes the organism in one sense and so that says okay well the organism is the unit of analysis and things lower our subgroups yes the beta cells in the in the pancreas have a niche that's surrounded by other types of cells and yes the oocyte is supported by other kinds of cells but still the organism is the unit of selection and so that gives um some people a reason to think that that is the default and that anything bigger than an organism is collective behavior anything smaller than the organism is molecular physiology or some different field but some of the implications of taking this multi-scale proposal seriously are that we still get the richness of collective behavior of lateral interactions within a level as well as the hierarchical layering and that combination of hierarchical layering with lateral interactions um goes by various names including a heteroarchival system from the social insect literature and when we start thinking about things like the whole genome like the genome encoding a developmental trajectory in relationship with an environment and a microbiome pretty interesting stuff and it leads to question what the unit of selection is is it the organism is it the kin group is it the kin group and the extended phenotype and one of the ways that we at least delay the question while deepening our understanding is by just observing and by modeling in the most integrative framework that we can imagine and then many of these other things fall out many of the relationships between higher and lower levels fall out but just because there's a relationship between a higher and a lower level spatially or laterally doesn't mean that one level or the other is the one where things are actually happening it's actually happening across all the levels okay oh yeah i mean i think that there's a there's a perspective that's being opened up here that really allows us to move beyond you know these kind of simplistic and empirically inadequate models like homo economicus like this idea that what human beings are really doing at the end of the day is optimizing their individual expected utility uh when they're performing tasks i mean it it turns out that humans act in a way that's deeply social we events a fairness psychology not nash equilibrium uh type so there's an interesting work in comparative ethology that puts uh chimpanzees against humans in a variety of economic tasks like the ultimatum game and yeah it turns out that humans just don't operate according to the principles of classical economics uh we don't zero in on nash equilibria it turns out that classical economics is actually pretty good at predicting the behavior of chimps though and so uh it's probably the case that humans in particular are subject to forms of social and cultural selection like forms of let's let's use the word you know it's group

selection right and you know this might have been spooky and hard to explain a while back but with the nested markov blanket formalism i think we're finally getting to a point where we can talk about group selection in a way that's not just slippery uh and in a way that doesn't reduce everything being selected to as you're saying dan uh to the genes yeah um two quick points on that first with the homo economicus it debases us twice it says your behavior is fundamentally economic one strike and two you're not even that good at it you make errors or you have these cognitive biases that lead you to make false positive investments or false negative instead we could just take a step back and observe how we see people interacting in a phylogenetic context just like you were describing and maybe that is how we are which is that we are pro-social and it's not an error correction on an error correction on a debasement it's actually we can step into the richness of us as social creatures rather than thinking of sociality as being recovered after we've already you know paid off everybody um and then also the question about group and kin selection and all these different levels of selection debate first off in the last several years through the work of akasha and others it has been described that some of these mathematical equations are actually formally identical so there's an isomorphism and it's like you can model it different ways but you're gonna get to some of the same consequences um and we can think about the affordances of a single cellular bacteria let's say now a single cellular bacteria's niche may also involve metabolites from conspecifics and social interactions like quorum sensing um that's an example where the replicator is still the single bacteria however there's collective behavior like quorum sensing and then the major transitions in evolutionary individuality so like the prokaryote plus another uh something to make a eukaryote eukaryotes coming together to make multicellular organisms and multicellular insects becoming the eusocial organisms it's not based upon you know formica economicus it's not the ants making an economic trade-off it's that when their affordances change in the context of a niche you end up seeing an evolutionary trajectory that favors obligate cooperation at that scale and precisely precisely it sidesteps a lot of these how would i feel if i were an aunt who didn't get to reproduce type questions which are really uh sidetracks relative to what is happening yeah and you know i think uh it's doing a lot of damage if we're being real about it i mean you know these these economic models are not just flawed theoretically and empirically inadequate they're also driving investment um and you know the the current state of the economy i think which is you know despite what you might hear from propaganda like not not so great right now i mean it's uh it it i think in part it has to do with the flawed way in which we're understanding you know human economic behavior what we

really need to be doing is fostering a move towards like you know value or affordance-based economies yeah or instead of how can we give everybody enough u.s coin so that they can buy food another way to phrase that is how could we reduce people's uncertainty about their next meal and maybe sometimes it looks this way sometimes it looks that way but if we were focused on reducing physiological surprise rather than as a proxy trying to increase some value on a spreadsheet thinking that people would then um just acquire the foods it's an interesting idea if we pivoted from these reward-based framings of political or economic issues to uncertainty based issues because it's often what you hear people needing help with is not knowing where the next place they're going to sleep or get their meal from not i don't have enough to get a meal i don't know where the meal is coming from so it's actually interestingly an uncertainty question even in that case oh yeah absolutely here's our next slide another quote from the paper is we hypothesize that the central mechanisms whereby cultural affordances are acquired especially conventional context involving affordances consists in looping our feedback relations between shared intentionality and shared attention in this light one can view social norms and conventions as devices to reduce mutual uncertainty that is constantly with a free energy framework as entropy minimizing devices one must know what is in the minds of others such as what one would see and how one would interpret another's action generally and in context in order to make a successful inference both explicit content involving or implicit correlational inferences about other agents in each simulation situation sorry so the question here was how do narratives reduce uncertainty how does conversation that's a great question um i mean you know you all did uh our paper on narratives a few weeks ago so i'm sure you have some elements so to answer that already but i mean that the fundamental idea i think uh is the and also i don't want to take up too much room if other people have more interesting things to say please uh by all means you know jump in but i think you know narratives do this in two main ways uh following the two main senses of narrative that we've outlined by a narrative you might understand uh some kind of internal cognitive shama right so like when you tell tell the story of your life or whatever and like from that point of view what narratives allow is for a kind of a novel kind of useful uh re-representation of lived events that you've already undergone so it's a way of kind of introducing new structure and meaning into the flow of events that that have happened to you but by narrative you might also mean a shared story so literally a cultural artifact that is produced through a storytelling practice and uh you know from from a niche construction point of view these are shared elements of a common that we interact with and effectively uh internalize the structure of such that we're we're then able to uh

you know redeploy the ideas concepts values and so on that are present in the structure of that narrative to structure our social life if i had in a nutshell i think those are the two main functions yep um and please anyone on team com raise your hand if you want to bring in the paper uh the idea of that narrative as almost being embodied in our evolutionary priors reminds me of the hero's journey and the the narrative structure of the hero's journey it's not about the hero tying and untying their shoes it's about a story arc which goes from a to z and that does many things it compresses the past into a distilled representation so that you don't need to remember every single detail of the past but rather you can remember the action-relevant sections of the past like the bird told me something relevant you don't need to remember how many feathers were on the bird it's enough to know that in a certain space and time you had an interaction where the bird told you go down this path not that one and then the symbolism as well and so words are symbols these are things we'll return to as well and so the narratives are like paths of symbols whether strings of words um or evocative uh patterns yeah stephen yeah and this also probably changes a bit in terms of how indigenous cultures work with story and legend and animalism and unknowns in the environment which they were tuning to and the way that we've kind of codified our narratives these kind of grand narratives which rather than it being about the land which indigenous culture tends to be about the land and timeless qualities that are enfolding you get this kind of you know you mentioned about the past as an entity that's in a timeline where we have a sense of self in the timeline which is actually kind of a more modern way of thinking about things so in some ways we've actually designed an understanding of how we fit into the world because our niche has changed in the way you know you were talking about because we've constructed our own way of understanding what our sort of human world is and attune to now we're kind of we're even redefining the way we see our history and the way we see narratives because narratives are these broader temporal depths which can stretch and then stories have become kind of commodified whereas in indigenous culture and story and land and us are much more kind of in the present in this kind of unfolded kind of cycle nature provides rather than saying what our calendars or our watches provide yeah two thoughts on that one is the distinction between kronos and kairos so the chronometer is the clock that's tick tick tick tick tick that's the regularities and the decimal system and kairos means timeliness and so it relates to dinner time versus 6 p.m those are very different things one is an exact time independent of an embodied experience and the other one is actually the experience independent of the clock time so that's important and then also just to highlight um the pragmatism of taking an action-oriented

approach or the active inference approach to thinking about our modern life as team com we often find it helpful when beginning and closing a meeting like bookends to almost have narrative review just restating syntactically here's who's here here's what we intend to get out of this interaction or here are the goals of the meeting in previous frameworks like getting things done is technically an action-oriented framework but not in the inactive embodied sense it's sort of in a productivity homo economicus framework and so you get convergent evolution of sometimes the recommendations like make a to-do list or restate the goals at the beginning of the meeting but rather than thinking about these as simply tasks to be achieved it actually helps us realign at an increasingly fine scale with the work that needs to be done and that is a full stack alignment from the syntactic level the details and the nitty gritty of the work that does need to get done up through the narrative where are we at this moment and what alphas does this project relate to that's from the systems engineering framework all the way up to the highest level which is what is our group doing what is our company doing what problems are we trying to solve and these ground floor or pet house level questions depending on how you think about it are foundational and we return again and again to these questions about what our purpose is and that's man's search for meaning that's the whole task and that doesn't really end so by framing the problem not just as a set of uh popping in tasks to the stack and then finishing them we can think about this as this ongoing process of realignment of the details of our sensory states to our desired regularities that we want to see in the any other thoughts on this slide cool all right so we are gonna have a few embodied affordance questions so here's a quote from the paper sports science provides an illustration of this tendency towards optimal grip studies of the dynamic interplay between a boxer's stance and position and the action possibilities available to them as a function of stance and position have shown that punching bags afford different kinds of strikes to boxers as a function of the distance between boxer and punching bag boxers tend to move their bodies to an optimal distance from the punching bag specifically one that affords the greatest variety of strikes so not the most powerful not the most rewarding not the most um unexpected to the other person in a theory of mind but actually the greatest variety from an embodied perspective of the actor this is a case of moving towards optimal grip so it's beyond the grasping now we're talking about stance when observing a painting we also move our bodies and our gazes in a way that maximizes our grip on the scene or details observed we might call such dynamic adaptive engagement with a field of affordances in rolling cycles of action perception skilled intentionality so the question here is what is being optimized by the body what do bodies do it's a good question there's probably

a couple of answers and we can maybe go reveal half of the hint which is how do predictions play into this optimization so what is the body optimizing um doesn't need to be the end-all be-all answer but what is something the body optimizes and then well i mean a lot of my own work has been about since writing this paper has been about figuring out exactly what this means uh but you know if the free energy principle holds then it's not just brains that are doing this free energy minimization business it's really every level of biological self-organization and there it really just means the same thing across all levels which means that you have a desired distribution over your data and you know actions acting on the environment is generated such that this preferred data distribution is brought about what changes at each scale is what the relevant data is what the relevant set points are and what the mechanisms are that bring about the preferred data distribution right so uh you know from a cellular point of view uh you know the markov blanket of a cell is effectively its membrane uh where the sensory states are all the receptor surfaces of the cell and the active states are like the cytoskeleton and other mechanisms that allow the cell to move about and to effectively generate the patterns of data that it expects to encounter but this very same story can be told at basically every level of like i was saying a biological self-organization where like from a from an interoceptive point of view uh what the body is doing is trying to keep some of its core homeostatic variables within uh viable ranges right so uh you know core body temperature might be maintained using such a mechanism where when you register a deviation from 36 and a half degrees you know you might have like a kind of low level uh autonomic kind of reaction you know you'll start shivering and you'll get goosebumps and so on uh but you know for uh animals like ourselves that have repertoires of cultural behavior well you might put on a parka you might turn your air conditioning on there are a bunch of other solutions to the same issue and i mean i think this is really the strength of the free energy principle i alluded to this last week uh i i at the time of writing this i sort of had an idea how this would work but it took the following four years of my life to really figure out what that entailed you know along with our our collaborators uh carl fristen and everyone else if i can add one point on this into how the predictions play into the so there's the classical control theory thermostat temperature drops and then the body can engage in the interoceptive affordances like uncoupling in the mitochondria or shivering or the behavioral and the uh cultural affordances like wearing a jacket but how about if you were planning a trip to somewhere that was cold so there's nothing in your local environment that's telling you that your temperature is dropping in fact you could be in arizona sweating but you're packing for somewhere that's cold and so traditional theories

of semiotics might be challenged by that they could say well you saw um the cold city and you looked at the website and so you made some sort of connection and you keep on doing this back propagation of symbols which is a valid account but also we can think about what the body's doing is optimizing its thermal range it's minimizing surprise about excursions from its thermal range that's tolerable and then it's acting based upon its prediction and so similarly there's substances or states where your body's prediction it's now cast prediction or its future prediction can be wrong and so you can have people who are shivering even though their body temperature is high in the case of the flu or something like that so it's really the interaction of the prediction and the affordances not necessarily the instantaneous however the instantaneous is often the most at hand um sasha um this is quite funny but this metaphor reminded me of um just kind of the stereotypical like when you go out your mother tells you to bring a jacket and that's because like the older caretaker role thinks farther ahead into future states and the younger child role um they're just thinking about right now or where they're going and they're not thinking to this possible future state they're not planning for that um possibility so i just thought that was a nice parallel to this whole uh jacket thermoregulation um how far ahead we're willing to plan to reduce um variability or uncertainty in our and differences potentially in risk tolerance too even say i i know it's going to be cold mom i just don't want to risk losing my jacket or i want to be cold or something that or i don't want to listen to you because whatever oh sasha i think you're putting your your finger on precisely like the core kind of political economic issue that we're dealing with as a species right now which is that the problems that we're dealing with are occurring at a scale that that is more or less invisible uh to the generative models that we've evolved uh in our heads effectively right so uh you know climate uh climate change global warming uh these are processes that unfold over decades and uh that require large scale responses uh you know probably to address this you know in any way decently like a kind of multinational coordinated response uh but our brains are meant to deal with phenomena spanning from a few seconds to you know maybe a year in in more traditional societies when you're like dealing with the season cycle and everything uh and so there's a huge gap there and you know i think the the disconnect between the kind of scientific consensus and uh you know at least in the us the kind of lay consensus on say climate change has to do with the difference in temporal depth of the models that we're looking at so as scientists we we deal with these like models that in in the case of like astrophysics allows us to increase by like you know orders and orders of magnitude for like the the scope of our predictions and uh and so on but then there's this gap you know uh between our our intuitive

understanding of the world and physics and this also connects back to the world that we live in you know uh what kind of world we share ultimately um yeah so all of that stuff yeah yes nice um stephen yeah i think i agree with what's being said and we've got into this world at the moment where we we're very focused on what we want and how we can achieve that but in our bodies might be when we're putting on the coat or something we're meeting our needs as an organism so that the heat and in complex adaptive systems for development they're looking at like need want and love to have happen and like the want is what we might you know and we tend to focus on this instrumental want but that can change within like this construct but our bodies we tune into this kind of deeper feeling of what we need like we can infer that and connect to that and i think that is um that's partly what active inference sort of helps us go back to okay when we're starting to sit into that because indigenous cultures look at that thing of seven generations and it's not like they plan for seven generations but they sense what's needed so they're kind of working at that kind of you know organic level one one way also that the temporality becomes embodied in our language in active through language is which words don't have plurals in english now this isn't like a hard and fast rule but oil fish sheep there's a lot of words where it's just like it's air it's just there's no airs how many errors are here it's just air is around us so implicitly one shouldn't have to worry about depleting it right well wrong because of course you can overfish and so it really bridges us to asking about how could we construct our actions today so that it influences the way that we want the future to be just to complete this slide the last question is beyond physical grip where does this idea of optimal grip apply and a really nice visualization was from yelle broomberg's and eric wrightville's paper on self-organization free energy minimization and optimal grip on a field of affordances which was cited in ramstead's paper and so here we see three representations of fields of affordances so these are like blocks that you could pick up they're blocks that you could sit on maybe blocks that you could carve a sculpture out of there are things that you can do stuff with and depending on the affordance it's gonna be a different one and the normal on the top in a we see that there's multiple different kinds of affordances of different types there's a bunch of yellow ones a bunch of red ones and a bunch of blue ones and maybe they have differential salience or they're adequate for different uses like a taller or a shorter person then in b and c we see two sort of hyperbolic and simplified but still didactic representations of alternate states so in b which is depression all of the field of affordances look less interesting they all look relatively drab and uh sort of not uh fun and in ocd we see that there's a massive highlighting of one of these affordances at almost the expense of completely blocking the

perception of all of these other and so i find this to be a really helpful framework because it helps highlight the commonalities between for example different people who are in a depressed state or different representations of obsession on different topics it could be rumination in thought it could be a specific motor behavior but when we think about it like this we can actually go from this representation backwards and say okay well could we change our perspective so that yes the red square is still really big but also we can see at least just for now just see these other affordances or could we make one of those gray boxes a little bit more exciting so i i just like this as a representation and it shows that this grip concept and the affordance idea is really transferable to different areas springberg eric grievell julian kaverstein don't necessarily get enough credit for this whole skilled intentionality framework that they've developed uh you know i i think it's it's some of the connective tissue that really allowed us to build the cultural affordances framework i really really like the way that they connect the kind of inactive emphasis on you know action orientation and embodied kind of coping with an with the ecological psychology literature on affordances and so on i mean uh we were discussing this last week their use of affordance is also not i think consonant with the kind of classical gibsonian sense of affordance uh where you know and i i mean the common sense and use of affordance i think in in academia at least also doesn't really resonate with what gibson meant recall that what gibson meant really was that uh all the information that an organism needs to interact with its environment is already available in its sensory array as it moves through its environment such you know an internal reconstruction of the environment wasn't all that necessary because you could just directly interact with the environment so a lot of these commitments are kind of falling out especially like in our use of the of the construct in the free energy principle framework like we're explicitly reintroducing notions like affordance sorry like uh inference uh you know like reconstructing the set of causes that led to the the sensory stimulation uh but i think uh you know some of the people who make uh the the concept palatable and usable in things like uh sports sciences and so on are these people pushing the skill intentionality framework like it's a it's a very useful framework especially i don't know like uh stephen i'd be curious to hear what you think about this as someone who's coming more at this from the practice and i think the concepts there are more readily usable than some of the more like formal stuff coming from active inference and so on and yet it's all kind of part of the same conceptual framework yeah i i agree i mean this is where it gets complex because you've got quite a radical form of affordances being thought of and intentionality and you've got obviously the active inference complexity but they're like

they talk about social practice theory in um as as quite a in environmental science work now a lot so they're trying to look at things through what's the social practices and the social norms that kind of align people i think it's the same kind of thing that you're talking about here it gives it gives an actual plausible mechanism so it doesn't jump from like this is what's happening in my body and then suddenly there's some theoretical framework which is somehow completely separate it's like okay it makes sense that i'm i'm enacting putting my coat on i've been acting looking at this painting and trying to understand it i'm now enacting looking at this painting like i used to work in them in graphic reproduction so when you look you know you're not looking at the painting anymore you're looking for the colors in it or is it going to reproduce or i'm going to zoom in on it with photoshop so it's like the practice of engaging with the environment and then like what's that fitted into if it's going to go into a publication so what's the intentionality what's the kind of scales of practices um and i think that i think what you're saying it makes sense i don't know if the two have quite managed to join up yet to make it fully practice ready because to try and explain it but i think what maybe that's where it just has to be used in practical applications in like an activity and maybe the people doing the activity don't need to know all the stuff that's going on behind the under the hood if that makes sense um so yep nice well i want to come to some of the other ideas in our other sections so this is section on ontologies and culture so here are two nice representations one of people working together all around the world and the other one of the skeleton that we all know and love of active inference the question is how do ontologies relate to culture and then if a person wanted to address that question with a raised hand or the second question is what kinds of tools are needed to work with ontologies and narratives so i'll start with one of language as an ontology scheme for the regularities and the objects that are around us so it's one of the cliches that x culture has y words for z concept you know oh this culture has 22 ways to describe this and of course you can always add adjectives to describe a variety of different kinds of snow or sand or whatever it is but it becomes deeper through time and there becomes specialized words that really capture and can finesse as an as a handleable as a grippable concept different types of sand snow whatever it happens to be trees and then can then be built into larger systems or sentences maxwell um yeah so i mean i i think what you just said was very interesting um i just want to clarify a bit what we meant by uh ontologies in this context um so i mean we're drawing on uh the ontological turn material uh that's coming out of anthropology in large part uh apologies to steven uh some of this material is delusion uh i mean you know i i also tend to share your your reservations about uh some of the uh stuff

coming out of delos and stuff but uh so the ontological turn fundamentally is saying something like well what we should be doing is talking about the worlds in which people inhabit in the way that they inhabit it right rather than just saying what we have as scientists as knowledge and what other people have is mere belief uh what we should be doing is talking about like the really the worlds in which people live and and the ontologies or like the categories of types of being that people are kind of bringing on board um so you know we we don't i think we're at odds with some of the people in the um the ontological term because we take a meta-bayesian approach to this which is to say well we all operate with beliefs the scientific worldview is also a set of beliefs that comes with its own ontology and whatever so as long as it's an even playing field and there's no like kind of disparity uh between you know the the respect with to that we're according to different kinds of knowledges then you know we're sort of all operating with our ontologies and in the context here in this paper what we mean more specifically is things like uh like ontologies of persons right like so uh for those of us in the american and particularly north american context you know racism and colorism our major issues and you know it just is the case that young children even even children that hail from minority groups internalize the categories of personhood uh of the dominant uh political group right so you know the the well-known you know uh experiments on children suggesting that you know children from racialized communities tend to prefer uh dolls for example that look like white people or whatever or dolls that uh are consonant with their own kind of ethnic uh you know outward appearance uh and so the yeah what what we're talking about then is like the categories of p you know uh different kinds of people uh different kinds of places uh all of these kind of concepts and norms and frameworks that so all this is an ontology it's it's a way of cutting up your sensory stream effectively of parsing your phenomenology and what we're suggesting is that this is acquired mainly through immersive participation in practices so you don't have to be explicitly told for example the first time you go to a fancy restaurant not to jump up and down on the the tables humans are exquisitely sensitive to context and we're able to just dynamically extract the norms of the situation well at least most neurotypical humans are able to do this on the fly without instruction and so this is where the ontology stuff at least comes in in this paper yeah alejandra yeah um well actually i was thinking about this um idea about the landscape affordances and the shared attention and how this shared attention is determining which expectations will be encoded in in this hierarchical predictive uh processing so i was wondering and also in terms of these scientologies related to culture um about social networks um it's kind of weird but yeah this is idiot it's very um like um

importantly in my mind how these um new ways of interacting um are changing it's it's a kind of a massive joint attention uh i i don't know how to to to say it but yeah yeah like a massive joint attention a kind of extended culture that is manipulating us um to what to attend so um in in in some ways is changing um what we believe that is relevant and um manipulating us um our attention and what to expect but in a massive way um and i think but we are not conscious about this um um the the effect that is having on on our um beliefs and as as you were saying maxwell uh this um racism and social preferences i think that um i don't i don't know if i'm explaining myself but this this massive joint attention um is is kind of affecting our our grip the way we grip with society in um unexpected ways i don't know yeah that makes total sense to me uh i really think that this is like a really key issue uh i mean you know the it's unfortunate but uh you know prejudicial behavior and like hate crimes have been on the rise steadily since you know the trump administration took power and uh i mean i think it speaks to the the power of uh you know the power of the of the media effectively and of shared images and shared narratives you know um i mean personally i think a lot of the issue comes from the fact uh you know older generations uh weren't taught to be information skeptic right uh so i mean you know uh growing up now as a 30-something when i was a kid and internet was like starting out you know what you learned was don't trust everything you read on the internet so there's the i think one of the the is that uh communities form around different types of epistemic standards effectively um and i mean if you sort of look at the split the cultural split the the increasingly deepening kind of cultural split uh between you know more conservative and more uh liberal uh populations in the u.s for example uh i think what you see is really a split in modes of adequate justification um you know so it's it's stunning to me how the same kind of fact base can lead to completely opposite interpretations uh in this kind of but unfortunately you know the uh the fact that you know uh you have like very hateful groups that now spread uh messages uh you know on a medium that then is shared i mean it really takes the it takes the uh joseph henrik uh puts this well uh in in his great book the the secret of our success he says uh essentially the same things that make us capable of incredible feats of coordination you know and that have allowed us to you know like build large-scale cities and develop modern medicine uh the same capacity for intense cooperation is also what makes us docile and what makes us crave totalitarian rulers uh we we just like to be very nice and agreeable you know and and if if everyone kind of decides that some group is going to be the scapegoat well you know there's a tendency to go along with that and you know if we're if our hypothesis is correct and there's also a tendency to just acquire unthinkingly download the

kind of information that's presented to us in immersive contexts then it's not surprising that you know if these messages are spreading online you know in very highly immersive kind of tunnel vision things like youtube channels that's just like re-link similar content i mean like yeah i don't know if this is an entirely coherent message but i do deeply resonate with what you were saying alejandra i think this is uh this is a huge set of issues that we need to address and it a lot of my own research for the next several years is going to be directed in precisely this direction uh like countering understanding and countering the spread of these narratives and like just modeling social dynamics using these tools more broadly yeah exactly as well sorry then just to add something um in terms of like your personal narrative your own story um how how is affected um i i can see it in in adolescence or in child's they are just expecting alike and that's meaning so meaningful to to um minimizing prediction error about yourself so uh yeah it's like you have like thousands or hundreds of people just paying attention about your picture and and if you receive 10 likes or 100 likes um what this like um the way that the self is gripping to to this um that's an interesting point there's a weird kind of self-evidencing i guess yeah it's very sad but i guess that it's really tragic when you think about it right we off sourced we off sourced our sovereignty of attention to algorithms and those are driven by exploit and in our sense of self-worth right so from an active inference point of view likes and retweets and shares and whatever are evidence for your existence right for for a model of you having worth in a social environment uh that's actually i think worth at least the paper if you want to write that up alejandra like send me a dm and we'll write that up as a paper it sounds it's a great idea um sasha uh yeah i just wanted to add on to that um a comment about uh don't trust the um and i i really like the phrase that you used alejandra uh the massive joint attention um and then it seems like social media um is kind of able to bypass that filter of don't trust the internet but these are my friends this isn't the internet these um the social network somehow is able to pass through that filter that this is a group of people that we trust that we feel that we trust more than the internet as this um greater entity well they don't call it anti-social media they don't call it physical distancing they say social distance these words really matter as far as slipping in the back door semantically with what um is uh being communicated in the end so yes um to pick up on what sasha was saying i think uh the internet's epistemic status is very much up in the air like you know i i think i'm i'm always a bit naively optimistic uh and so i like to think that people are critical about what they read on the internet but i i think it was a joke somewhere i forget who said it but it's something like you know when i was growing up my mother told me don't get into a car with strangers and like now i literally summon from my

internet having mobile device to pick me up and to bring new places uh like the uh i i mean as some of you are pointing out it's it's sort of the connective tissue of a lot of relations especially post coval a lot of my friendships for example have been pursued over the uh so that's actually a very interesting point maybe the fact that this information is being presented through our friends that are sharing this uh information does tend to increase its precision for us and make it more salient 100 it's why the word of mouth recommendation of a product is the highest form it's why influencers try to be personable there's so much to that angle and using the pre-existing and the possible social networks to leverage uh the spreading of memes that might be just purely corrosive not even for or against one side just pure disinformation or pure things that uh are divisive let me just run through a few of our latter questions so we can get some more neurons firing connect some more ideas and stabilize these concepts so in this quote which um i will not read it's just about how through cultural practices and i found a really nice image that i thought tied together ontologies with culture so this is an image from the ontotext.com on the bottom there and it writes ontologies do not only introduce a shareable and reusable knowledge representation but can also add new knowledge about the domain so i had a few questions here about ontologies and that could be in that anthropological term the the anthropology or the ontological turn as maxwell was describing it or in the more computational sense like gene ontologies the structure of words that we use to describe genes so ontology in that sense means like a formal knowledge structure and about the attention and relationship between attention and ontology between ontologies and development and then especially for this group how do ontologies and active inference interact and how does that actually add knowledge to a situation rather than being a pure descriptor well i mean in part you don't grasp the world in a naked way right like we grasp it through our conceptual categories uh through the salience structures that are encoded in our conceptual categories in large part and uh yeah i mean the uh the point of having an ontology is to kind of you know fix the frame problem in a sense to kind of say okay what out of all these elements that could be monopolizing my attention at a given moment you know who are the relevant interlocutors that are deserving of my what are the salient aspects of the scene that require my processing yeah and in general what you know just in in the broad sense what uh is kind of the otherwise you're just you know dealing with a sensory stream which is all fun and good but like so the ontology is what you bring to bear on the sensory stream to kind of parse things out from it yep and i i like where you took that it was exactly what i was thinking about like how many tennis balls can you fit in your hand maybe three maybe you have a large hand four tennis

balls or something but you have a bag and you can cluster the tennis balls and you could pick up a bag with 100 tennis balls so that's the grip physical metaphor is the tools the affordances that cluster allow you to grip more and so similarly we can use ontologies of ideas and use that to get a grip on really high level and then we'll pick up the double edged sword though right because uh i mean you know this is the problem of using priors in general is that you know you're starting from a point that might not be the right one and that might actually prevent you from uh you know so ontologies are the reason why we have you know the capacity to interact with each other for example as peers you know because we have these categories uh you are my peers and my friends and we can interact but you know ontologies are also responsible for racist behavior uh for sexism uh you know if i if my ontology says women are inferior uh to men or the people with dark skin but i don't believe any of that just to be clear but like if that if that is my ontology then i'm bound to treat uh others in a way that reflects those beliefs uh so i mean this is why they're so important to culture i think because they kind of it's it's not unrelated to this formal definition that you were discussing dan like it gives you the kind of general tracts that you'll follow in your treatment of others uh so then it becomes interesting to you know use anthropological methods for example to see what are the ontologies that people are actually bringing to bear uh you know and to use that physical carrying metaphor so you go okay well i can hold three eggs in my hand i should have a bag with 500 eggs because i can pick it up well now you have none or you have a mess and so over simplification through any number of mechanisms within a value structure simply leads to false positives and false negatives and another way that we can bring in the free energy principle is the phrasing of free energy as model accuracy penalized by complexity and in some ways the extreme uh simple model let's say a one category model like all people are people or two or three category model and so do the models get more accurate in terms of reducing surprise as we include more and more parameters and then in the extreme case we have everyone as unique which is an bookshelf you know end piece to contrast with this everybody's the same and so somewhere in between is where we're actually going to end up and so how do we help bring these implicit generative models of the world and our priors in into being shaped and designed and uh personally i think we never escape the value question and it's one thing to come together with somebody who has a different perspective and it's why it's so important to highlight coming together across differences because there's just no amount of cultural or genetic or whatever homogeneity that is going to prevent conflict and so instead we can have a coming together story about working together across differences rather than wedging tiny differences and that

story itself is in a trade-off with essentially gaslighting and telling people that suffering is just part of how things are so there's so many trade-offs that exist um especially when we're talking about issues that matter most i mean we're not talking about species definitions in foxes we're talking about our own lives and our world in our ecosystem and so yes they're touchy topics and it's so important to be clear about what is and also what we want to see and so i i think there's so much that can be explored and expanded on here i i like that you brought in complexity minus accuracy uh because in a sense the you know this is this is a little bit a slightly looser way of thinking about it but in general you want as little boxes as possible right uh boxes i mean if you listen to buddhist wisdom for example boxes are the source of suffering right we we cling to our boxes we put things in boxes and and it makes it it makes us feel comfortable uh but really this is the source of suffering uh you know the this tendency to only be comfortable with things when they're nicely packaged in these kind of nice binary categories and this is why people uh get so like uncomfortable uh when these binaries get challenged you know by like gender fluidity and transness and these kinds of things on the flip side if we didn't have boxes at all we would die right uh you know you need to be able to parse that sensation as hunger uh in order to generate like you know appropriate survival generating uh behavior so i think that's like the the the fundamental paradox of human existence is both that we need uh boxes to survive but that they also the source of much you know just suffering uh both that were inflicting to ourselves by forcing these kind of conceptual straitjackets onto ourselves but also because of the social wrongs that that can be uh the result of you know using the wrong ontology yep and cognitive diversity means differences in risk tolerance it means of course different perspectives unique perspectives and also um maybe on that explore exploit or on that model accuracy versus complexity some people will accept a more accurate model from their perspective that is also a lot more complex whereas somebody else is the coefficient of how much they're penalizing a complex model is different so they're going to come to a different conclusion and it's not as simple as right versus wrong in that situation it's model preference and could we take a step back could we reframe and understand how we're all part of the ensemble that's learning it would be nice let's go through a few of these last topics on communication and semiosis or meaning generation so here is from figure one of the paper which we discussed last time as far as what it was describing but here i want to connect it to one of the quotes they write conventions need not be explicitly formulated as rules and may instead originate in the actor's engagement with local backgrounds over time that is from non-contentful developmental experiences learning or participation in social and cultural

practices to operate with conventional affordances agents must have shared sets of expectations we must know what others expect us to expect so yes we're hitting the same idea from figure one but there's a lot we're kind of spiraling another layer and coming back to this idea where we're in a remote group right now and so we are getting our expectations about what to expect and all these things not only are they going to be very different person to person because our local worlds are quite different but there's this algorithmic level that influences what we expect and it changes on the inputs that we receive and reducing our surprise about the sensory inputs is one of the drivers of behavior and so how does what we see in the news feed end up recapitulating the things that we will draw our attention to and that's sort of this self-radicalization process that can happen is attention is brought to something and then it continues to basically percolate upwards as people pay more and more attention and are more and more cautious or more and more aware of something that it's hard to take a step back from any scientific worldview and say that one is better or the other because although i'd love to hear a workaround it's hard to get a purely secular perspective because someone could say i'm just pursuing my accuracy by doing something that's aberrant to you and it's difficult to see purely within the framework nor does there have to be i think what free energy and active inference and focusing on the communication dynamics allows us to do is let our values be values let our communication be communication let our be a behavior and uh talk about what we expect using ways that we want to communicate and if people don't want to use non-violent communication other people may or may not choose to communicate with them but it's um it change how we communicate what we expect when there's not only that computational intervention but also a lot of um you know coming together of different perspectives it's one of the challenges of our day and then also just this cultural iceberg was a nice image some of the the words are small in there but the quote on here is joint attention is usually understood as occurring in a diet of two people or between agents in direct interactional spheres of communication gaze following finger-pointing or other verbal or non-verbal cues to address more complex social situations it is useful to revise current sociocognitive models of joint attention to encompass fundamentally triadic situations in which the third is the socially constituted niche of affordances supported by local ontologies and abilities so before you have the questions just some of these in on the iceberg above the iceberg we have dress folklore language fine arts holidays food and below we have family values we have gender roles um pride concept of justice so this and there's different versions of the iceberg i just thought this one had a lot of nice ideas my questions here were what is the

third member of a social interaction and how can active inference help us design culture if i can jump in quickly of it's sort of so a human once in culture it is never alone in a sense uh we're always constantly seeing things through the eyes of others it starts off as what would mommy want me to do in this situation and it it just gen i mean it's it's remarkable Canadians are really good at this uh we uh there were i i'd have to pull up the studies but there are there are studies that show that Canadians even at night you know when there are no policemen will stop at a red light and just wait it out you know and and go even if there are no agents that could penalize them uh i mean it's really remarkable the extent to which we we internalize uh the these kinds of norms and they end up then directing uh the way that we interact with uh with others yeah any other thoughts on this or we can just it's it's something for pondering because we have the explicative portion of our experience which is that which comes to our conscious awareness that which is stated verbally but on the input side subconscious input and on the generative models that can't be expressed in words sometimes they can be expressed through art or through action but the whole iceberg is there and so it's just i really like that once you're in culture you're never alone and indeed we are never alone because to be aware is to be communicating it's almost like the uh cog ergo sum it's like i communicate therefore we exist something like that um in this last page on um semiosis which i think is an topic the authors wrote however cultural symbols and signs are usually poly semis that means having many or multiple meanings and their interpretation depends on context it's contextual our hypothesis is that feedback loops mediating shared attention and shared intentionality are the principal mechanism whereby cultural especially conventional affordances are acquired so conventional is fun i think in this setting because it alludes to conventional like normal like it's just the conventional approach it's the normal approach but also based upon convention and convention in that sense means both the coming together like it's a convention of the knife sharpeners or something but also it's a joint it's agreed upon so it has many senses and it shows how normal and agreed upon often are related that's one symbol and so here was a few graphics that i found on a nice the dsproject.com so on the left is what is known as the piercing triad by c.s pierce and that's the sign the interpretant and the object and on the right is um sarsour who is another semiotician though may have used different words and here we see that there's the signifier and the signified and then the referent and so i like this because it shows that there's multiple ontologies even when we're talking about sense making about meaning generation there's multiple ways we can parse that and do we choose to see it as a triad or do we choose to see it as a binary with another binary like the sr and the sd are one whole and

then the frf is another you know is it a pair of binaries or is it just an integral triad so that's one way where there's multiple frameworks we can use just to talk about meaning generation and then this image is a representation by those authors about unlimited semiosis and this is sort of looking through time how that triad evolves and so i just thought how does this relate to active inference to that unfolding idea to this notion that our meaning generation our sense-making process is ongoing with respect to the symbols but we can modify those symbols as well so how can we maybe use active inference at least to grasp what are the outlines of this problem some of it i think was addressed in uh later papers for example uh i forget which active stream but when we talked about is the free energy principle a formal theory of semantics and it relates to that um i think it would probably be cool at almost the end of our time just to switch into panel mode and just see if anyone had any last thoughts that they wanted to talk about any last questions or where are you going with what we've discussed today what do you think would be a good question to leave people on or a good next step for us to take i thought this was a really interesting and useful exercise i'm super grateful to all of you for this continued uh this continued attention uh to the work that we've done uh yeah i i told the i told the team earlier but i've i've blocked off these uh mornings to participate regularly um so i mean i i i'm very satisfied with this i think it's very cool and uh i think we've raised a lot of issues that could be pursued as papers uh yeah particularly this uh issue that you were raising alejandra about uh social networks and the way that they tie into uh attentional dynamics i think uh should be the target of much more uh discussion uh yeah i mean i i thought everyone had very very interesting uh things to say so thank you here's one angle on that is an evolutionary prior is that your social networks are constituted of people who are not adversarial with you most of the people who you're engaged with whether or not they believe the exact same thing you have to be relaxed enough around them to sleep for example and so the the deepest prior is if we're relating then we're going to be to some broad extent in agreement okay now we know from active inference and niche construction that the two ways that you can um deal with uncertainty or with unexpected sensory inputs is by changing the environment that's niche construction or by changing your internal model that's inference so what happens when you're in a social contrived situation it says oh but your friend posted this okay so it's coming in through the friend door and it's uh something that for my current value system at time t i think of is not chill there's all kinds of things not being chill but it just isn't and the two ways that i can deal with that are change my model to be more like that so i make that less surprising some type of hateful speech i get numb to it or i come to embody it actually to reduce my

surprise on the inference side or i can get doubled down in my beliefs and try to reduce my uncertainty or the surprise from those sensory cues by getting into a debate i mean think about how much mental energy gets burnt on these sort of debates and people trying to change each other's mind online and there's something beautiful about people sharing their differences in idea but when that whole discussion is being basically overviewed by an algorithm that doesn't exactly have those people's mental health or the alignment of different cultures in mind not exactly yeah not not exactly you know it's correlated no it it can get off the rails and so that deep evolutionary prior isn't something that can just be unlearned like you were speaking about earlier maxwell and so when our deepest prior and our intergenerational ones get taken advantage of it's like these slow waves and they can't quite cope with these very fast frequencies and so the options are to slow it down that's like the rko futurist you know oh well in the future we'll return to a more heterogeneous uh organization of people so slow down these high frequency waves or come to a better way that we can predict and communicate about them i mean there's a lot more that we could think about there but it just it's uh yeah there's yeah uh steven and then alejandra well i like you were saying these ways that we think about what is happening um is really important in terms of how we construct our conversations but i think there's also the where and the where we've taken our attention to be able to have our conversations so when you're locked in they call it context collapse when you're on like facebook because you have no context you have no story you have no so one of the things with complex adaptive systems the workers cynthia kurtz and david snowden they work with um stories for small micro stories so you fight a hundred a thousand and start to make sense of them like attribute them with um titles and metadata and actually work with story because when you're working with story each story if it's kept raw so it's not an opinion they call it it's like it's kind of maintains the internal complexity of it so you can still reinterpret it but once you've got an opinion an opinion is collapsed down you can't really reinterpret an opinion so we're kind of trapped in these kind of interpretive kind of inductive deductive spaces which is great if it's a case of like i have the right answer or you have the right answer so then we can like debate it but the trouble is what if we don't know and no one wants to admit they don't know um can we work with something more complex maybe that's a rare question as a how question not a what question a knife alejandra um yeah and and i i totally agree and also taking what uh sasha said that the idea that we are talking with our friends and yeah that's like the like our conscious um internal belief they are our friends but we actually not have any idea how these platforms work um how these platforms are

manipulating our attention so we are communicating with others but we have something between this communication that we don't know actually how is working and um i don't know i i was thinking that um they are kind of playing in terms of active inference to constantly increase our prediction never um and you have to engage with with this platform in order reduce or minimize prediction error in terms of emotions or um the way i react to other um giving a or um sharing a post so um i don't know this idea that we are interacting with friends or we're um close people but we don't know what's happening in between sasha um yeah i think that that's a great point um to touch on that we don't really understand what's happening uh this hidden state between um us and the person that we're trying to communicate with and how that's being manipulated and while that's that's a totally valid critique of social media platforms um we could say the same thing for like telephone communication or even telegram um the people who were sending the message shouldn't necessarily understand what was happening um in in that process and yet uh they felt or they they would trust that it was um the integrity of the message was maintained or that it wasn't uh nothing was lost in that process or manipulated but so it's been kind of like a steady a steady march of uh technol technological advances that are more and more removed from what we think we're doing in this process of communication yep the neutral claim is that context directs attention so if there's a bunch of arrows pointing to something even though it's not in the exact object it will draw your attention to it or something that's placed on top of a pyramid is what your attention is supposed to be drawn towards well in the telegram case not the app the morse code version the context is that you trust that western union or whomever isn't going to be garbling your message and the context is who it's from and so you can actually unpack it as if that person were just telling you but when it's a stranger and it's like engaging the gas and the break on is this an adversary or not and the context is constrained so you can only see their name or a fake internet name and a picture and then is it a surprise that given a picture and what somebody says somebody's gonna unpack it in potentially the least charitable way based upon identity or based upon a typo based upon some sort whereas in person we just look at each and we can have more of this human sense we can have the context and that helps us understand and then beyond the context cues when we can actually feel like we're part of a shared society then we want to help people who are in need in our local worlds and there's only one world so it is our local world spaceship earth so there's there's so many angles and the question would just what are those on ramps and then how is it all going to actually play out any final thoughts though because this was an awesome discussion great follow-up and there's we could have talked for longer on just

four point two three four five but next week we'll move to five and so if there's no further thoughts i really appreciate everybody coming together thanks for the attention to those participants and also to our listeners we'll always be even weeks later accepting input on what papers what topics what questions would be cool to discuss and so yes thank you it's been a great